

Division Decoded

Every graded set, every week — 5 weeks, 25 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Two Kinds of Division

Facts first, then split the dividend into friendly pieces.

✦ Warm-Up 12 ITEMS • RECALL

1 $42 \div 6$

2 $56 \div 8$

3 $63 \div 9$

4 $36 \div 4$

5 $72 \div 8$

6 $45 \div 5$

7 $12 \div 3$

8 $20 \div 4$

9 $18 \div 2$

10 $25 \div 5$

11 $30 \div 10$

12 $16 \div 4$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $84 \div 4$ ($80 \div 4 = 20$, then $4 \div 4 = 1$)
.....

14. $96 \div 6$ ($60 \div 6 = 10$, then $36 \div 6 = 6$)
.....

15. $75 \div 5$ ($50 \div 5 = 10$, then $25 \div 5 = 5$)
.....

16. $90 \div 6$ ($60 \div 6 = 10$, then $30 \div 6 = 5$)
.....

17. $78 \div 3$ ($60 \div 3 = 20$, then $18 \div 3 = 6$)
.....

18. $56 \div 7$
.....

19. $48 \div 6$
.....

20. $81 \div 9$
.....

 Challenge NOT GRADED • REASON IT OUT

21. Area 96, one side 6. Other side? ($96 \div 6$)

22. 72 pencils, 8 per box. Boxes? ($72 \div 8$)

23. $144 \div 12$

24. $132 \div 11$

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

The Missing Side

Area and one side are given. Break the area up and hunt the other side.

✦ Warm-Up 12 ITEMS • RECALL

1 $24 \div 3$

2 $54 \div 6$

3 $81 \div 9$

4 $32 \div 8$

5 $49 \div 7$

6 $60 \div 5$

7 $14 \div 2$

8 $21 \div 3$

9 $28 \div 4$

10 $35 \div 5$

11 $40 \div 8$

12 $27 \div 9$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $128 \div 4$ ($120 \div 4 = 30$, then $8 \div 4 = 2$)
.....

14. $156 \div 6$ ($120 \div 6 = 20$, then $36 \div 6 = 6$)
.....

15. $144 \div 3$ ($120 \div 3 = 40$, then $24 \div 3 = 8$)
.....

16. $175 \div 5$ ($150 \div 5 = 30$, then $25 \div 5 = 5$)
.....

17. $168 \div 7$ ($140 \div 7 = 20$, then $28 \div 7 = 4$)
.....

18. $84 \div 7$
.....

19. $96 \div 8$
.....

20. $108 \div 9$
.....

21. $66 \div 6$
.....

22. $77 \div 7$
.....



Challenge

NOT GRADED • REASON IT OUT

23. Area 252, one side 7. Other side? ($210 \div 7 = 30$, then $42 \div 7 = 6$)

24. Bigger answer: $144 \div 6$ or $168 \div 8$? Type it. ($144 \div 6 = 24$ and $168 \div 8 = 21$)

25. $156 \div 12$

26. $196 \div 14$



Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE



Error journal

ONE SENTENCE PER MISTAKE

When It Doesn't Fit

Quotient and remainder are two different answers. Read which one is being asked for.

Warm-Up 6 ITEMS • RECALL

1 $10 \div 5$

2 $24 \div 6$

3 $36 \div 6$

4 $45 \div 5$

5 $32 \div 8$

6 $50 \div 10$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

7. $17 \div 5 \rightarrow$ quotient (3 fives is 15, and 2 are left) _____

8. $94 \div 4 \rightarrow$ quotient ($80 \div 4 = 20$, then $12 \div 4 = 3$) _____

9. $137 \div 6 \rightarrow$ quotient ($120 \div 6 = 20$, then $12 \div 6 = 2$) _____

10. $92 \div 4$ _____

11. $78 \div 6$ _____

12. $85 \div 5$ _____

13. $126 \div 9$ _____

14. $115 \div 5$ _____

★

Challenge

NOT GRADED · REASON IT OUT

15. $17 \div 5 \rightarrow$ remainder
-
16. $23 \div 4 \rightarrow$ remainder
-
17. $38 \div 6 \rightarrow$ remainder
-
18. $50 \div 7 \rightarrow$ remainder
-
19. $29 \div 3 \rightarrow$ remainder
-
20. $45 \div 8 \rightarrow$ remainder
-
21. $94 \div 4 \rightarrow$ remainder ($4 \times 23 = 92$)
-
22. $137 \div 6 \rightarrow$ remainder ($6 \times 22 = 132$)
-
23. 26 kids, vans hold 6. Vans needed? (Four vans carry 24 – two kids still need a seat)
-
24. 26 cookies for 6 friends. Whole cookies each? (This time the leftover stays on the plate)
-
25. $288 \div 12$
-
26. $384 \div 16$
-

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Chunks Get Tidy

Bigger dividends. Pull out the biggest easy chunk first, then finish.

✦ Warm-Up 12 ITEMS • RECALL

1 $120 \div 4$

2 $150 \div 5$

3 $240 \div 6$

4 $350 \div 7$

5 $180 \div 9$

6 $160 \div 8$

7 $9 \div 3$

8 $15 \div 5$

9 $22 \div 2$

10 $33 \div 3$

11 $44 \div 4$

12 $55 \div 5$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $372 \div 3$ ($300 \div 3 = 100$, $60 \div 3 = 20$, $12 \div 3 = 4$)
.....

14. $465 \div 5$ ($450 \div 5 = 90$, then $15 \div 5 = 3$)
.....

15. $728 \div 8$ ($720 \div 8 = 90$, then $8 \div 8 = 1$)
.....

16. $594 \div 6$ ($540 \div 6 = 90$, then $54 \div 6 = 9$)
.....

17. $952 \div 7$ ($700 \div 7 = 100$, $210 \div 7 = 30$, $42 \div 7 = 6$)
.....

18. $144 \div 6$
.....

19. $168 \div 8$
.....

20. $175 \div 7$
.....

21. $192 \div 6$
.....

22. $225 \div 9$
.....

Remainder Race & Puzzles

Enrichment day. Every leftover needs a decision — say which one you made.

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

✦

Warm-Up

2 ITEMS • RECALL

1

18 ÷ 6

2

42 ÷ 7

◆

Core

GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

3.

91 ÷ 7

4.

120 ÷ 8

Week 1 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

1.1 • Two Kinds of Division

OUT OF 20

Warm-Up • 1-12 • 7 | 7 | 7 | 9 | 9 | 9 | 4 | 5 | 9 | 5 | 3 | 4

Core • 13-20 • 21 | 16 | 15 | 15 | 26 | 8 | 8 | 9

Challenge • 21-24 • 16 | 9 | 12 | 12

1.2 • The Missing Side

OUT OF 22

Warm-Up • 1-12 • 8 | 9 | 9 | 4 | 7 | 12 | 7 | 7 | 7 | 7 | 5 | 3

Core • 13-22 • 32 | 26 | 48 | 35 | 24 | 12 | 12 | 12 | 11 | 11

Challenge • 23-26 • 36 | 24 | 13 | 14

1.3 • When It Doesn't Fit

OUT OF 14

Warm-Up • 1-6 • 2 | 4 | 6 | 9 | 4 | 5

Core • 7-14 • 3 | 23 | 22 | 23 | 13 | 17 | 14 | 23

Challenge • 15-26 • 2 | 3 | 2 | 1 | 2 | 5 | 2 | 5 | 5 | 4 | 24 | 24

1.4 • Chunks Get Tidy

OUT OF 22

Warm-Up • 1-12 • 30 | 30 | 40 | 50 | 20 | 20 | 3 | 3 | 11 | 11 | 11 | 11

Core • 13-22 • 124 | 93 | 91 | 99 | 136 | 24 | 21 | 25 | 32 | 25

Challenge • 23-26 • 211 | 1 | 63 | 72

Fri • Remainder Race & Puzzles

OUT OF 4

Warm-Up • 1-2 • 3 | 6

Core • 3-4 • 13 | 15

Challenge • 5-15 • 14 | 2 | 13 | 84 | 7 | 6 | 56 | 12 | 36 | 25 | 125

Round It Up

Vans, boats and buses. Two leftovers still force one more of everything.

Warm-Up 7 ITEMS • RECALL

1 $20 \div 5$

2 9 people, vans of 4 — vans needed

3 12 people, vans of 4

4 13 people, vans of 4

5 7 people, boats of 3

6 $11 \div 3$ — the whole part

7 $26 \div 5$ — the whole part

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

8. 127 children, buses of 30 — buses needed

.....

9. 98 books, boxes of 12 — boxes needed

.....

10. 53 eggs, cartons of 6 — cartons needed

.....

11. 200 seats, rows of 18 — rows needed

.....

12. 75 guests, tables of 8 — tables needed

.....

13. $83 \div 9$ — the whole part

.....

14. $94 \div 7$ — the whole part

.....

Drop It

Whole ones each. The leftover just stays on the plate.

Warm-Up 9 ITEMS • RECALL

1 $26 \div 6$ — whole cookies each

2 $26 \div 6$ — left on the plate

3 $17 \div 5$ — whole each

4 $30 \div 7$ — whole each

5 $30 \div 7$ — leftover

6 $19 \div 4$ — whole each

7 $20 \div 3$ — the whole part

8 $14 \div 4$ — the whole part

9 $25 \div 6$ — the whole part

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

10. \$50 shared by 8 — whole dollars each

.....

11. \$50 shared by 8 — dollars left

.....

12. $145 \div 12$ — whole each

.....

13. $145 \div 12$ — leftover

.....

14. $200 \div 30$ — whole each

.....

15. $100 \div 7$ — the whole part

.....

16. $137 \div 9$ — the whole part

.....

★ Challenge NOT GRADED • REASON IT OUT

17. 90 cm of ribbon into 14 cm bows — whole bows

18. That ribbon — cm wasted

19. $20 \div 3$ — the remainder

20. $14 \div 4$ — the remainder

21. $9 \div 2$ — the remainder

22. $100 \div 7$ — the remainder

23. $137 \div 9$ — the remainder

24. $88 \div 6$ — the remainder

25. $500 \div 24$ — the whole part

26. $500 \div 24$ — the remainder

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal ONE SENTENCE PER MISTAKE

Share It Out

Cut the leftovers up and the remainder becomes a fraction.

Warm-Up 7 ITEMS • RECALL

1 $9 \div 2$ — whole part

2 7 kids share 15 apples — each gets

3 ... and left over

4 4 kids share 10 — each

5 ... left over

6 3 kids share 12 — each

7 5 kids share 22 — each

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

8. $17 \div 4$ — type the fraction part as a/b

.....

9. \$45 shared by 4 — dollars each

.....

10. 6 vans hold 8, 50 kids — vans needed

.....

11. Boxes of 12, 100 items — full boxes

.....

12. ... items left over

.....

13. 9 seats a row, 60 people — rows needed

.....

14. Bags of 6, 45 marbles — full bags

.....

The Remainder Is the Answer

Clocks and repeating patterns. Sometimes the leftover is the whole point.

Warm-Up 7 ITEMS • RECALL

1 Days in a week

2 $8 \div 2$

3 $12 \div 4$

4 $15 \div 3$

5 $18 \div 9$

6 $24 \div 8$

7 $30 \div 6$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

8. Pattern red, blue, green repeating. The 10th bead — type 1, 2 or 3 for its place in the group

.....

9. It is 9 o'clock. 25 hours later, the hour shown

.....

10. $105 \div 5$

.....

11. $128 \div 4$

.....

12. $147 \div 7$

.....

13. $162 \div 6$

.....

14. $189 \div 9$

.....

★

Challenge

NOT GRADED · REASON IT OUT

15. $10 \div 3$ — the remainder
-
16. $14 \div 7$ — the remainder
-
17. $25 \div 4$ — the remainder
-
18. $9 \div 7$ — the remainder
-
19. $20 \div 6$ — the remainder
-
20. $100 \div 7$ — the remainder
-
21. $50 \div 12$ — the remainder
-
22. $365 \div 7$ — the remainder
-
23. A 4-colour pattern. The 63rd item — its place in the group of 4
-
24. $1,000 \div 7$ — the remainder
-
25. $456 \div 8$
-
26. $729 \div 9$
-

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Snack Run Planning

Real quantities, real people. Every leftover decision written down.

✦ Warm-Up 4 ITEMS • RECALL

1 24 crackers between 4 — each

2 20 grapes between 3 — whole each

3 $16 \div 8$

4 $21 \div 7$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. A pack of 18 between 5 — whole each
.....

6. A pack of 18 between 5 — leftover
.....

7. $98 \div 7$
.....

8. $117 \div 9$
.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 • Round It Up

OUT OF 14

Warm-Up • 1-7 • 4 | 3 | 3 | 4 | 3 | 3 | 5

Core • 8-14 • 5 | 9 | 9 | 12 | 10 | 9 | 13

Challenge • 15-26 • 1 | 12 | 23 | 1 | 2 | 2 | 4 | 5 | 3 | 3 | 10 | 1

2.2 • Drop It

OUT OF 16

Warm-Up • 1-9 • 4 | 2 | 3 | 4 | 2 | 4 | 6 | 3 | 4

Core • 10-16 • 6 | 2 | 12 | 1 | 6 | 14 | 15

Challenge • 17-26 • 6 | 6 | 2 | 2 | 1 | 2 | 2 | 4 | 20 | 20

2.3 • Share It Out

OUT OF 14

Warm-Up • 1-7 • 4 | 2 | 1 | 2 | 2 | 4 | 4

Core • 8-14 • $\frac{1}{4}$ | 11.25 | 7 | 8 | 4 | 7 | 7

Challenge • 15-26 • 0.5 | 0.25 | 4.5 | 3.5 | 0.75 | 6.5 | 3.25 | 4.4 | $\frac{3}{4}$ | 12.5 | 7 | 13

2.4 • The Remainder Is the Answer

OUT OF 14

Warm-Up • 1-7 • 7 | 4 | 3 | 5 | 2 | 3 | 5

Core • 8-14 • 1 | 10 | 21 | 32 | 21 | 27 | 21

Challenge • 15-26 • 1 | 0 | 1 | 2 | 2 | 2 | 2 | 1 | 3 | 6 | 57 | 81

Fri • Snack Run Planning

OUT OF 8

Warm-Up • 1-4 • 6 | 6 | 2 | 3

Core • 5-8 • 3 | 3 | 14 | 13

Challenge • 9-14 • 2.4 | 5 | 8 | 64 | 69 | 143

Under the Bar

Partial quotients folded into the standard layout, line by line.

✦ Warm-Up 9 ITEMS • RECALL

1 $60 \div 3$

2 $80 \div 4$

3 $90 \div 3$

4 $48 \div 4$

5 $66 \div 6$

6 $84 \div 4$

7 $40 \div 4$

8 $120 \div 4$

9 $150 \div 5$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

10. $96 \div 4$

11. $126 \div 3$

12. $175 \div 5$

13. $192 \div 6$

14. $133 \div 7$

15. $168 \div 4$

16. $216 \div 6$

17. $294 \div 7$

18. $312 \div 8$

19. $378 \div 9$

Three-Digit Dividends

Estimate before you start. Every time, out loud.

✦ Warm-Up 12 ITEMS • RECALL

1 $300 \div 3$

2 $400 \div 4$

3 $360 \div 6$

4 $250 \div 5$

5 $280 \div 7$

6 $540 \div 9$

7 $100 \div 10$

8 $200 \div 10$

9 $300 \div 100$

10 $500 \div 5$

11 $600 \div 6$

12 $800 \div 4$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $372 \div 3$

14. $456 \div 4$

15. $595 \div 5$

16. $728 \div 8$

17. $651 \div 7$

18. $420 \div 6$

19. $560 \div 8$

20. $630 \div 9$

21. $720 \div 8$

22. $810 \div 9$



Zeros in the Quotient

The place nobody remembers to hold. It catches everybody once.

Warm-Up 12 ITEMS • RECALL

1 $60 \div 6$

2 $90 \div 9$

3 $200 \div 2$

4 $606 \div 6$

5 $404 \div 4$

6 $800 \div 8$

7 $22 \div 2$

8 $36 \div 3$

9 $52 \div 4$

10 $70 \div 5$

11 $90 \div 6$

12 $112 \div 7$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $618 \div 6$

14. $721 \div 7$

15. $915 \div 3$

16. $832 \div 8$

17. $525 \div 5$

18. $243 \div 3$

19. $364 \div 4$

20. $455 \div 5$

21. $546 \div 6$

22. $637 \div 7$

Check by Multiplying

Quotient $\times$ divisor + remainder = dividend. Self-marking from here on.

✦ Warm-Up 10 ITEMS • RECALL

1 7×8

2 $56 \div 8$

3 $9 \times 6 + 2$

4 $48 \div 4$

5 $6 \div 6$

6 $0 \div 5$

7 $7 \div 1$

8 $11 \div 11$

9 $9 \div 1$

10 $0 \div 12$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. $106 \div 6$ — the quotient
.....

12. $330 \div 3$
.....

13. $440 \div 4$
.....

14. $550 \div 5$
.....

15. $606 \div 6$
.....

16. $707 \div 7$
.....

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

✦ Warm-Up

4 ITEMS • RECALL

1 42 ÷ 6

2 90 ÷ 3

3 27 ÷ 3

4 48 ÷ 8

◆ Core

GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. 372 ÷ 3

6. 618 ÷ 6

7. 127 children, buses of 30 — buses needed

8. 252 ÷ 6

9. 336 ÷ 7

★ Challenge NOT GRADED • REASON IT OUT

10. $26 \div 4$ as a decimal

11. $100 \div 7$ — the remainder

12. A quotient of 23 remainder 5 with divisor 9 — the dividend

13. $1440 \div 12$

14. $2500 \div 25$

15. $999 \div 3$

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 • Under the Bar

OUT OF 19

Warm-Up • 1-9 • 20 | 20 | 30 | 12 | 11 | 21 | 10 | 30 | 30

Core • 10-19 • 24 | 42 | 35 | 32 | 19 | 42 | 36 | 42 | 39 | 42

Challenge • 20-23 • 36 | 49 | 144 | 243

3.2 • Three-Digit Dividends

OUT OF 22

Warm-Up • 1-12 • 100 | 100 | 60 | 50 | 40 | 60 | 10 | 20 | 3 | 100 | 100 | 200

Core • 13-22 • 124 | 114 | 119 | 91 | 93 | 70 | 70 | 70 | 90 | 90

Challenge • 23-26 • 117 | 144 | 300 | 600

3.3 • Zeros in the Quotient

OUT OF 22

Warm-Up • 1-12 • 10 | 10 | 100 | 101 | 101 | 100 | 11 | 12 | 13 | 14 | 15 | 16

Core • 13-22 • 103 | 103 | 305 | 104 | 105 | 81 | 91 | 91 | 91 | 91

Challenge • 23-26 • 510 | 1006 | 91 | 91

3.4 • Check by Multiplying

OUT OF 16

Warm-Up • 1-10 • 56 | 7 | 56 | 12 | 1 | 0 | 7 | 1 | 9 | 0

Core • 11-16 • 17 | 110 | 110 | 110 | 101 | 101

Challenge • 17-26 • 48 | 58 | 372 | 106 | 4 | 728 | 212 | 567 | 101 | 1010

Fri • Mid-Unit Quiz

OUT OF 11

Warm-Up • 1-4 • 7 | 30 | 9 | 6

Core • 5-11 • 124 | 103 | 5 | 6.5 | 2 | 42 | 48

Challenge • 12-15 • 212 | 120 | 100 | 333

Tests for 2, 5 and 10

The last digit is enough on its own. Say why.

Warm-Up 12 ITEMS • RECALL

1 Is 46 divisible by 2 — yes or no

2 Is 45 divisible by 5

3 Is 45 divisible by 2

4 Is 70 divisible by 10

5 Is 75 divisible by 10

6 Is 128 divisible by 2

7 $14 \div 7$

8 $24 \div 3$

9 $32 \div 4$

10 $54 \div 6$

11 $64 \div 8$

12 $45 \div 9$

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

13. Is 3,450 divisible by 10 — yes or no
.....

14. Is 1,275 divisible by 5
.....

15. Is 2,346 divisible by 2
.....

16. Numbers below 50 divisible by both 2 and 5
.....

17. Is 999 divisible by 5
.....

18. $238 \div 7$
.....

19. $272 \div 8$
.....

20. $306 \div 9$
.....

21. $204 \div 6$
.....

22. $170 \div 5$
.....



Working space



Error journal

Tests for 3 and 9

Add the digits — then find out why that ridiculous trick works.

Warm-Up 12 ITEMS • RECALL

1 Digit sum of 24

2 Is 24 divisible by 3 — yes or no

3 Digit sum of 45

4 Is 45 divisible by 9

5 Digit sum of 17

6 Is 17 divisible by 3

7 Half of 20

8 Half of 36

9 A quarter of 20

10 A third of 21

11 A fifth of 25

12 Half of 50

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

13. Digit sum of 738

.....

14. Is 738 divisible by 9 — yes or no

.....

15. Is 1,234 divisible by 3

.....

16. Is 5,346 divisible by 9

.....

17. Digit sum of 9,999

.....

18. A quarter of 96

.....

19. A third of 111

.....

20. A fifth of 145

.....

21. An eighth of 96

.....

22. A sixth of 144

.....



Working space



Error journal

Four-Digit Dividends

Same method, longer bar. Do not flinch.

Warm-Up 12 ITEMS • RECALL

1 $1,000 \div 2$

2 $2,000 \div 4$

3 $3,000 \div 3$

4 $4,800 \div 8$

5 $5,600 \div 7$

6 $1,200 \div 6$

7 \$20 shared by 4

8 \$30 shared by 5

9 \$18 shared by 3

10 \$40 shared by 8

11 \$60 shared by 6

12 \$100 shared by 10

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $3,472 \div 8$

14. $4,536 \div 6$

15. $5,432 \div 4$

16. $6,741 \div 3$

17. $8,127 \div 9$

18. \$144 shared by 12

19. \$175 shared by 7

20. \$210 shared by 6

21. \$252 shared by 9

22. \$180 shared by 15

★

Challenge

NOT GRADED · REASON IT OUT

23. 9,072 ÷ 8

24. 7,595 ÷ 5

25. \$1000 shared by 8

26. \$1440 shared by 24

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Two-Digit Divisors

Grade-five ceiling. Estimate, adjust, estimate again.

Warm-Up 12 ITEMS • RECALL

1 $40 \div 20$

2 $60 \div 30$

3 $90 \div 30$

4 $80 \div 20$

5 $120 \div 40$

6 $150 \div 50$

7 $12 \div 6$

8 $20 \div 5$

9 $28 \div 7$

10 $40 \div 10$

11 $36 \div 4$

12 $49 \div 7$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $84 \div 12$

14. $96 \div 16$

15. $224 \div 14$

16. $375 \div 25$

17. $391 \div 17$

18. $456 \div 6$

19. $532 \div 7$

20. $608 \div 8$

21. $684 \div 9$

22. $380 \div 5$



Perfect Numbers

$6 = 1 + 2 + 3$. Its factors add up to itself. Hunt the next one.

✦ Warm-Up 4 ITEMS • RECALL

1 Factors of 6 below 6 —
their sum

2 Factors of 8 below 8 —
their sum

3 $33 \div 11$

4 $60 \div 12$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Factors of 28 below 28 — their sum

.....

6. Factors of 12 below 12 — their sum

.....

7. $441 \div 7$

.....

8. $512 \div 4$

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Tests for 2, 5 and 10

OUT OF 22

Warm-Up • 1-12 • yes | yes | no | yes | no | yes | 2 | 8 | 8 | 9 | 8 | 5

Core • 13-22 • yes | yes | yes | 4 | no | 34 | 34 | 34 | 34 | 34

Challenge • 23-26 • 10 | 19 | 34 | 34

4.2 • Tests for 3 and 9

OUT OF 22

Warm-Up • 1-12 • 6 | yes | 9 | yes | 8 | no | 10 | 18 | 5 | 7 | 5 | 25

Core • 13-22 • 18 | yes | no | yes | 36 | 24 | 37 | 29 | 12 | 24

Challenge • 23-26 • 0 | 3 | 24 | 111

4.3 • Four-Digit Dividends

OUT OF 22

Warm-Up • 1-12 • 500 | 500 | 1000 | 600 | 800 | 200 | 5 | 6 | 6 | 5 | 10 | 10

Core • 13-22 • 434 | 756 | 1358 | 2247 | 903 | 12 | 25 | 35 | 28 | 12

Challenge • 23-26 • 1134 | 1519 | 125 | 60

4.4 • Two-Digit Divisors

OUT OF 22

Warm-Up • 1-12 • 2 | 2 | 3 | 4 | 3 | 3 | 2 | 4 | 4 | 4 | 9 | 7

Core • 13-22 • 7 | 6 | 16 | 15 | 23 | 76 | 76 | 76 | 76 | 76

Challenge • 23-26 • 23 | 27 | 76 | 76

Fri • Perfect Numbers

OUT OF 8

Warm-Up • 1-4 • 6 | 7 | 3 | 5

Core • 5-8 • 28 | 16 | 63 | 128

Challenge • 9-14 • 28 | 496 | 496 | 168 | 144 | 55

Missing-Digit Division

Reason backwards from the quotient to the digits under the bar.

✦ Warm-Up 12 ITEMS • RECALL

1 $\ ? \div 4 = 6$

2 $\ ? \div 7 = 8$

3 $36 \div ? = 9$

4 $81 \div ? = 9$

5 $\ ? \div 5 = 12$

6 $48 \div ? = 6$

7 $8 \div 4$

8 $18 \div 3$

9 $35 \div 7$

10 $48 \div 12$

11 $55 \div 5$

12 $72 \div 9$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $\ ? \div 6 = 103$

14. $372 \div ? = 124$

15. $\ ? \div 8 = 91$

16. $\ ? \div 12 = 15$

17. $595 \div ? = 119$

18. $126 \div 6$

19. $203 \div 7$

20. $288 \div 8$

21. $351 \div 9$

22. $425 \div 5$

 Challenge NOT GRADED • REASON IT OUT

23. A 3-digit number $\div 7 = 43$ exactly — the number

24. $? \div 9 = 47$ remainder 3

25. $1224 \div 8$

26. $1575 \div 25$

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Snack Run

Real quantities, real people, every leftover call written down.

Warm-Up 12 ITEMS • RECALL

1 30 crackers between 5 — each

2 24 grapes between 4 — each

3 18 between 6 — each

4 25 between 5 — each

5 40 between 8 — each

6 21 between 7 — each

7 $25 \div 5$

8 $42 \div 6$

9 $56 \div 8$

10 $63 \div 7$

11 $81 \div 9$

12 $100 \div 4$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. A 44-pack between 6 — whole each

.....

14. A 44-pack between 6 — leftover

.....

15. \$18 split 4 ways — dollars each

.....

16. You need 5 each for 9 people. Packs of 12 needed

.....

17. You need 5 each for 9 people. Packs of 12 needed — snacks left over

.....

18. $175 \div 5$

.....

19. $216 \div 4$

.....

20. $301 \div 7$

.....

21. $464 \div 8$

.....

22. $531 \div 9$

.....

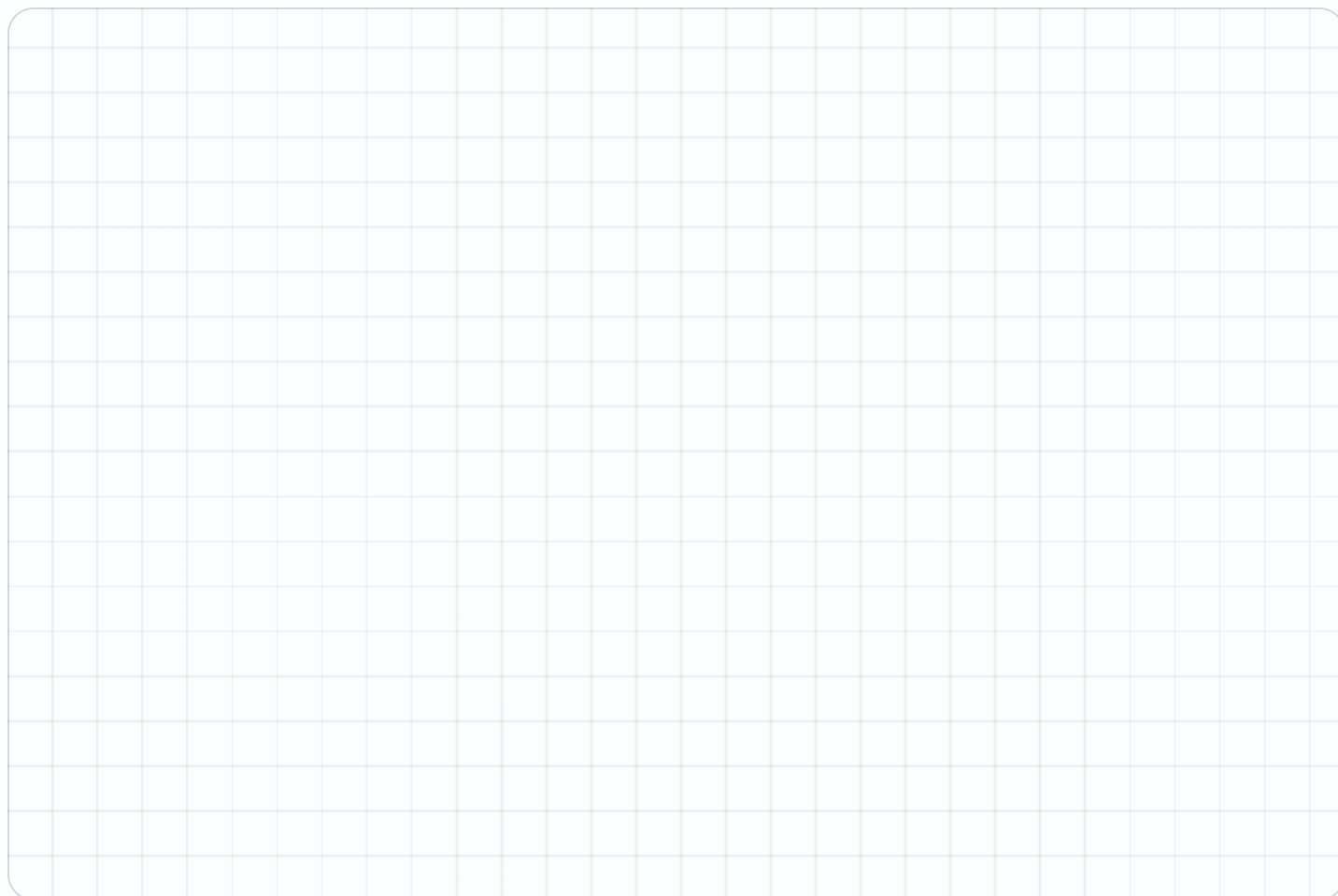
★ Challenge NOT GRADED • REASON IT OUT

23. \$25 buys packs at \$4 — whole packs

24. That purchase — dollars change

25. $2400 \div 16$

26. $3150 \div 45$

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Remainder Defence

One division, four right answers. Say which one the question wanted.

Warm-Up 10 ITEMS • RECALL

1 $27 \div 4$ — the whole part

2 27 children in cars of 4 — cars needed

3 27 cookies between 4 — whole each

4 $27 \div 4$ rounded down

5 $30 \div 5$

6 $44 \div 4$

7 $66 \div 6$

8 $77 \div 7$

9 $88 \div 8$

10 $99 \div 9$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. $53 \div 8$ — the whole part

.....

12. 53 people in vans of 8 — vans needed

.....

13. \$53 shared by 8 — whole dollars each

.....

14. $132 \div 4$

.....

15. $264 \div 8$

.....

16. $396 \div 12$

.....

17. $231 \div 7$

.....

18. $165 \div 5$

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Error Journal Sweep

Mixed review of the whole mission. Fix only what repeats.

Warm-Up 10 ITEMS • RECALL

1 $63 \div 9$

2 $120 \div 4$

3 Is 45 divisible by 9 — yes or no

4 $81 \div 9$

5 $200 \div 5$

6 $10 \div 2$

7 $16 \div 2$

8 $21 \div 3$

9 $32 \div 8$

10 $50 \div 5$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. $456 \div 4$

.....

12. $618 \div 6$

.....

13. $3,472 \div 8$

.....

14. $84 \div 12$

.....

15. $294 \div 6$

.....

16. $343 \div 7$

.....

17. $392 \div 8$

.....

18. $441 \div 9$

.....

19. $245 \div 5$

.....

Mission 02 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 4 ITEMS • RECALL

1 $48 \div 6$

2 $150 \div 5$

3 $24 \div 4$

4 $40 \div 5$

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. $372 \div 3$

6. $728 \div 8$

7. $915 \div 3$

8. $3,472 \div 8$

9. $96 \div 16$

10. Is 5,346 divisible by 9 — yes or no

11. 127 children, buses of 30 — buses needed

12. $156 \div 6$

13. $322 \div 7$

 Challenge NOT GRADED • REASON IT OUT

14. $26 \div 4$ as a decimal

15. $945 \div 35$

16. A quotient of 23 remainder 5 with divisor 9 — the dividend

17. $1728 \div 24$

18. $4356 \div 66$

19. $5000 \div 40$

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 5 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

5.1 • Missing-Digit Division

OUT OF 22

Warm-Up • 1-12 • 24 | 56 | 4 | 9 | 60 | 8 | 2 | 6 | 5 | 4 | 11 | 8

Core • 13-22 • 618 | 3 | 728 | 180 | 5 | 21 | 29 | 36 | 39 | 85

Challenge • 23-26 • 301 | 426 | 153 | 63

5.2 • Snack Run

OUT OF 22

Warm-Up • 1-12 • 6 | 6 | 3 | 5 | 5 | 3 | 5 | 7 | 7 | 9 | 9 | 25

Core • 13-22 • 7 | 2 | 4.5 | 4 | 3 | 35 | 54 | 43 | 58 | 59

Challenge • 23-26 • 6 | 1 | 150 | 70

5.3 • Remainder Defence

OUT OF 18

Warm-Up • 1-10 • 6 | 7 | 6 | 6 | 6 | 11 | 11 | 11 | 11 | 11

Core • 11-18 • 6 | 7 | 6 | 33 | 33 | 33 | 33 | 33

Challenge • 19-26 • 3 | 6.75 | 6.625 | 5 | 6 | 10 | 33 | 33

Thu • Error Journal Sweep

OUT OF 21

Warm-Up • 1-11 • 7 | 30 | yes | 2 | 9 | 40 | 5 | 8 | 7 | 4 | 10

Core • 12-21 • 114 | 103 | 434 | 7 | 2 | 49 | 49 | 49 | 49 | 49

Challenge • 22-25 • 23 | 426 | 49 | 49

Fri • Mission 02 Test

OUT OF 14

Warm-Up • 1-4 • 8 | 30 | 6 | 8

Core • 5-14 • 124 | 91 | 305 | 434 | 6 | yes | 5 | 6.5 | 26 | 46

Challenge • 15-19 • 27 | 212 | 72 | 66 | 125